Model Predictive Control for Stabilization of an Underactuated Two-Link Pendulum

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***Abstract*—This project studies a model predictive control (MPC) approach for stabilizing an underactuated planar two-link pendulum. The system consists of a pendulum with a single actuator at the first joint, while the second joint is passive. The control objective is to stabilize the equilibrium in which the first link hangs downward and the second link is inverted relative to the first link, while respecting actuator torque limits. A discrete-time prediction model is obtained by linearizing the pendulum dynamics around the desired equilibrium and discretizing the resulting continuous-time system. Based on this model, a finite-horizon MPC problem is formulated with quadratic stage and terminal costs, together with state and input constraints. Special attention is given to the stability analysis of the closed-loop system through the choice of terminal cost and terminal set. Numerical simulations are used to evaluate the controller performance and to compare the constrained MPC design with a local LQR controller. The results illustrate the effect of explicitly handling constraints and show the advantages of MPC for constrained local stabilization of the pendulum.**

I. Introduction

We consider a planar two-link pendulum with a single actuator at the first joint, while the second joint is passive. The equilibrium of interest is the configuration in which the first link hangs downward and the second link is inverted relative to the first link, as shown in Figure 1. This makes the system underactuated, since the second link cannot be controlled directly but only through the motion of the first link. The control objective is to stabilize this equilibrium while respecting the torque limits of the actuator. A common approach for this type of problem is to design a linear quadratic regulator (LQR) around the desired equilibrium, which can provide good local stabilization. However, LQR does not explicitly account for actuator limits and may therefore require torques that are not physically feasible. Model Predictive Control (MPC) is better suited to this problem because the torque limits can be included directly as constraints, while the controller also predicts the motion of the pendulum over a finite horizon. This is especially useful here, since the passive second link is influenced only indirectly through the first link, so the effect of the control input depends on how the motion develops over the next time steps.

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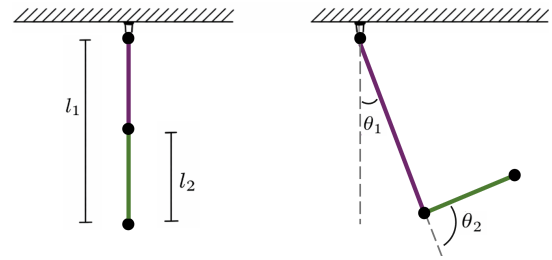

Fig. 1. Schematic configurations of the two-link pendulum. The left figure shows the considered equilibrium position, in which the first link hangs vertically downward and the second link is upright relative to the first link. The right figure shows a deviated configuration, described by the angles θ_1 and θ_2.

A. Nonlinear Dynamics of the Pendulum

The pendulum is modeled as a planar two-link system with a fixed base. The first link is connected to the base by an actuated revolute joint, while the second link is connected to the first link by a passive revolute joint. The two links are assumed to be massless, and the masses m_1 and m_2 are concentrated at the ends of the first and second link, respectively. The link lengths are denoted by l_1 and l_2, the gravitational acceleration by g, and the control input by u, which is the torque applied at the first joint only. In table I the parameters used can be found. An inertial reference frame N is used together with reference frames A_1 and A_2 attached to the first and second link. The generalized coordinates are chosen as θ_1, the angle of the first link with respect to the inertial frame, and θ_2, the angle of the second link relative to the first link. The corresponding angular velocities are $\dot{\theta}_1$ and $\dot{\theta}_2$. With this definition, the absolute angle of the second link is given by $\theta_1 + \theta_2$.

TABLE I
Parameters

Parameter	Value
l_1	0.1m
l_2	0.2m
m_1	0.1kg
m_2	0.3kg

The equations of motion are derived using Lagrange's method from the kinetic and potential energy of the system. If we define

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix},$$

the nonlinear equations of motion can be written in the compact form

$$M(q)\ddot{q} = f(q, \dot{q}, u) \tag{1}$$

where $M(q)$ is the mass matrix and $f(q, \dot{q}, u)$ contains the gravity, coupling, and input terms. The model is rewritten in first-order state-space form by defining the state vector as

$$x = \begin{bmatrix} \theta_1 & \theta_2 & \dot{\theta}_1 & \dot{\theta}_2 \end{bmatrix}^T.$$

The nonlinear dynamics are then written as

$$\dot{x} = F(x, u), \tag{2}$$

with

$$F(x, u) = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_1(x, u) \\ \ddot{\theta}_2(x, u) \end{bmatrix}. \tag{3}$$

B. Linearization around the Equilibrium

For controller design, the nonlinear model is linearized around the equilibrium of interest. This equilibrium corresponds to the configuration in which the first link hangs downward and the second link is inverted relative to the first link, with both angular velocities equal to zero and no applied torque. In terms of the state and input variables, the equilibrium is given by

$$x_e = \begin{bmatrix} 0 & \pi & 0 & 0 \end{bmatrix}^T, \qquad u_e = 0. \tag{4}$$

This linearization is justified because the controller is designed for local stabilization around this equilibrium. Since the pendulum is intended to remain close to the desired configuration, the linear model provides a reasonable approximation of the nonlinear dynamics in that region.

To describe the motion near this operating point, we introduce the deviation state, $\tilde{x} = x - x_e$, which describes the difference between the current state and the equilibrium state. The nonlinear state equation 2 is then linearized about (x_e, u_e), which gives the continuous-time linear model:

$$\dot{\tilde{x}} = A_c\tilde{x} + B_c u, \tag{5}$$

where the matrices A_c and B_c are obtained by differentiating the nonlinear state equation $F(x, u)$ 3 with respect to the state and input, and evaluating the result at the equilibrium point. The matrix A_c describes how the dynamics change for small deviations in the state, while B_c describes how the dynamics respond to small changes in the control input. The resulting linear model therefore approximates the nonlinear pendulum locally around the chosen equilibrium.

To verify that the linearized model is suitable for controller design, the controllability of the pair (A_c, B_c) is checked. This is done by forming the controllability matrix

$$\mathcal{C} = \begin{bmatrix} B_c & A_cB_c & A_c^2B_c & A_c^3B_c \end{bmatrix}.$$

For the linearized pendulum model, this matrix has full rank equal to 4, which shows that the linearized system is controllable around the chosen equilibrium. Therefore, all state variables can be influenced through the input torque.

C. Discrete-Time Prediction Model

Since MPC is implemented in discrete time, the continuous-time linearized model in (5) must be converted into a discrete-time model. The sampling time T_s is chosen using the eigenvalues of the continuous-time matrix A_c. Specifically, the largest eigenvalue magnitude is used as a rough measure of the fastest system dynamics, and the sampling frequency is set to twenty times this value. This results in a sampling time of $T_s \approx 0.02\,\mathrm{s}$. The continuous-time model is then discretized using zero-order ensuring the discretized system has the same reaction to the input as the continuous system.

Using the deviation variables introduced in the previous subsection, the final prediction model is written in standard discrete-time state-space form as

$$\tilde{x}_{k+1} = A_d\tilde{x}_k + B_d u_k, \qquad y_k = C_d\tilde{x}_k + D_d u_k. \tag{6}$$

Here, the state vector is

$$\tilde{x}_k = \begin{bmatrix} \theta_{1,k} \\ \theta_{2,k} - \pi \\ \dot{\theta}_{1,k} \\ \dot{\theta}_{2,k} \end{bmatrix} \in \mathbb{R}^4, \tag{7}$$

the input is the applied torque,

$$u_k \in \mathbb{R}, \tag{8}$$

and the output is chosen as the full state,

$$y_k = \tilde{x}_k \in \mathbb{R}^4. \tag{9}$$

This corresponds to full-state feedback, so that

$$C_d = I_4, \qquad D_d = 0.$$

The resulting discrete-time model therefore has state dimension of 4, input dimension of 1, and output dimension of 4.

II. Model predictive control design

The goal of the MPC design is to compute a control sequence that minimizes a quadratic cost while respecting both the limits of the physical actuator and the numerical validity of the linearized model. Using the linearized discrete-time model in Section I, the controller predicts the trajectory of the underactuated double pendulum over a finite horizon N. At each sampling instant, a constrained optimization problem is solved to ensure that the control action accounts for future coupling effects between the actuated and passive links.

Using the sampling time T_s introduced in the discretization step, the prediction horizon is chosen as $N = 20$, corresponding to a prediction window of $20T_s \approx 0.4$s. While this prediction window is relatively short in absolute time 10 prediction steps was found to be sufficient for the local stabilization problem considered here, which is shown in Section IV.

A. Constraints

The control problem is subject to several hard constraints that represent the physical limitations of the actuator and the numerical range in which the linearized model remains valid.

State constraints are introduced to keep the pendulum close to the equilibrium around which the model was linearized. Since the MPC controller is based on a local linear approximation of the nonlinear dynamics, large angular deviations would reduce the accuracy of the prediction model. Therefore, bounds are imposed on the first-link angle and on the deviation of the second link from its equilibrium position, namely $|\theta_{1,k}| \leq \theta_{1,\max}$ and $|\tilde{\theta}_{2,k}| \leq \tilde{\theta}_{2,\max}$ Here, $\theta_{1,\max}$ is chosen as a fixed bound, while $\tilde{\theta}_{2,\max}$ is taken equal to the initial deviation. This is not meant as a fixed physical limit, but as a study-dependent bound used to check whether the controller can recover the pendulum while keeping the passive-link motion within the local validity region of the linearized model. In compact form, these constraints can be written as $H\tilde{x}_k \leq h$, with:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \qquad h = \begin{bmatrix} \theta_{1,\max} \\ \theta_{1,\max} \\ \tilde{\theta}_{2,\max} \\ \tilde{\theta}_{2,\max} \end{bmatrix}. \tag{10}$$

We also include a control input constraint on the actuator torque $-u_{\max} \leq u_k \leq u_{\max}$, to represent the physical torque limits of the motor. This prevents the controller from demanding inputs that are not feasible in practice. The input constraints can be written compactly as $Gu_k \leq g$, with:

$$G = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad g = \begin{bmatrix} u_{\max} \\ u_{\max} \end{bmatrix}. \tag{11}$$

The numerical constraint values used in this implementation are summarized in Table II. They are discussed in section IV.

TABLE II
NUMERICAL CONSTRAINT VALUES

Parameter	Symbol	Value
Max. First Link Angle	$\theta_{1,\max}$	90°
Max. Second Link Deviation	$\tilde{\theta}_{2,\max}$	90°
Max. Actuator Torque	$u_{\max}$	1.5 Nm

B. Cost Function and Terminal Ingredients

The MPC objective is to minimize a quadratic cost that penalizes deviations from the equilibrium together with the applied control effort. The stage cost is chosen as

$$\ell(\tilde{x}, u) = \tilde{x}^\top Q \tilde{x} + u^\top R u \tag{12}$$

where $Q \succ 0$ and $R \succ 0$ are weighting matrices. We use $Q = \mathrm{diag}(q_{\theta_1}, q_{\tilde{\theta}_2}, q_{\dot{\theta}_1}, q_{\dot{\theta}_2})$ so that the angular deviations and angular velocities can be weighted independently. Larger weights on θ_1 and $\tilde{\theta}_2$ enforce tighter regulation of the two links around the equilibrium, while R is used to penalize large actuator torques. A relatively large weight was assigned to the second joint deviation, since the second link is passive and therefore harder to stabilize directly. The remaining weights were chosen to balance state regulation and control effort, so that the controller achieves fast stabilization without requiring unnecessarily aggressive torque inputs. The numerical values of Q and R, together with the tuning procedure, are discussed in Section IV.

To ensure stability and a close approximation of the infinite-horizon cost, we include a terminal cost

$$V_f(\tilde{x}) = \tilde{x}^\top P \tilde{x} \tag{13}$$

where $P \succ 0$ is obtained from the discrete-time algebraic Riccati equation (DARE) associated with the LQR design for the linearized system (A_d, B_d).

Finally, we include a terminal set constraint $\tilde{x}_N \in \mathbb{X}_f$. In this project, $\mathbb{X}_f$ is taken as an ellipsoidal set of the form $\mathbb{X}_f = \{\tilde{x} \in \mathbb{R}^4 \mid \tilde{x}^\top P \tilde{x} \leq \alpha\}$. Positive invariance of this set is ensured by choosing α such that, under the local LQR control law $u = K\tilde{x}$, the closed-loop state remains inside $\mathbb{X}_f$ and the state and input constraints are satisfied. The set is computed by finding the largest value of α for which these conditions hold, ensuring that the MPC problem remains feasible once the terminal set is reached.

Both a terminal cost and a terminal set are included to satisfy the standard stability conditions for constrained MPC, rather than omitting them and relying only on a longer prediction horizon. The specific computation of the LQR gain K and the determination of the maximum feasible value of α are detailed in Section III, where the stability and feasibility conditions are formally verified.

C. Finite-Horizon Optimization Problem

Using the discrete-time prediction model, the state and input constraints, the stage cost, and the terminal ingredients, the MPC controller is obtained by solving a finite-horizon optimization problem at each sampling instant. Let $u_N := \begin{bmatrix} u_0 & u_1 & \cdots & u_{N-1} \end{bmatrix}^\top \in \mathbb{R}^N$ denote the input sequence over the prediction horizon. For a given initial condition $\tilde{x}_0$, the optimization problem $P_N(\tilde{x}_0)$ is formulated as:

$$\begin{aligned} \min_{u_N} \quad & \sum_{k=0}^{N-1} \left(\tilde{x}_k^\top Q \tilde{x}_k + u_k^\top R u_k \right) + \tilde{x}_N^\top P \tilde{x}_N \\ \text{s.t.} \quad & \tilde{x}_0 = \tilde{x}_{init}, \\ & \tilde{x}_{k+1} = A_d \tilde{x}_k + B_d u_k, \quad k = 0, \ldots, N-1, \\ & H\tilde{x}_k \leq h, \qquad\qquad k = 0, \ldots, N-1, \\ & Gu_k \leq g, \qquad\qquad k = 0, \ldots, N-1, \\ & \tilde{x}_N \in \mathbb{X}_f. \end{aligned} \tag{14}$$

By applying only the first optimal input u_0^* and repeating the optimization at the next step, we achieve a receding horizon control law.

III. Asymptotic stability

In this section, the stability of the closed-loop MPC system is established using the standard constrained MPC stability theorem from the textbook [1]. The main result used is **Theorem 2.19**, which gives asymptotic stability of the origin provided Assumptions 2.2, 2.3, 2.14, and 2.17 are satisfied. In addition, **Theorem 2.21** is used to strengthen this conclusion to exponential stability, provided the corresponding Lyapunov bounds hold.

The analysis is carried out in deviation coordinates, so that the equilibrium of interest is shifted to the origin. We consider the discrete-time prediction model introduced in Section I, together with the quadratic stage cost, terminal cost, terminal control law, and terminal set defined in Section II.

A. Verification of Basic Assumptions

Assumption 2.2: This assumption requires continuity of the system dynamics and the cost functions. The discrete-time prediction model introduced in Section I is linear and therefore continuous. The stage cost in (12) and the terminal cost in (13) are quadratic functions and are therefore continuous as well. In addition, Assumption 2.2 requires the origin to be an equilibrium and the stage and terminal costs to be zero at the origin. Since the analysis is carried out in deviation coordinates, the equilibrium of interest is shifted to the origin, so that $f(0,0) = 0$. Also, $\ell(0,0) = 0$ and $V_f(0) = 0$. Hence, Assumption 2.2 is satisfied.

Positive definiteness of the stage cost: The stage cost is defined with $Q \succ 0$ and $R \succ 0$. It therefore follows that $\ell(\tilde{x}, u) > 0$ for all $(\tilde{x}, u) \neq (0,0)$, so the stage cost is positive definite. This is important because the standard stability result of Theorem 2.19 relies on the stage cost penalizing every nonzero deviation from the equilibrium.

Terminal control law existence: A local terminal control law exists through the discrete-time LQR design for the pair (A_d, B_d). In particular, solving the discrete-time algebraic Riccati equation yields the matrix P, from which the corresponding feedback gain K is obtained. The resulting local law $u = K\tilde{x}$ is used as the terminal controller in the stability analysis. Moreover, the controllability check of the linearized system, carried out in Section I, supports the use of this LQR design and therefore the existence of a suitable terminal feedback law.

Properties of the terminal cost: The terminal cost is chosen as $V_f(\tilde{x}) = \tilde{x}^\top P \tilde{x}$, where $P \succ 0$ is the solution of the discrete-time algebraic Riccati equation associated with the LQR design. Since P is positive definite, $V_f(\tilde{x})$ is also positive definite and satisfies $V_f(0) = 0$. Also, under the local terminal control law $u = K\tilde{x}$, the terminal cost decreases along the closed-loop dynamics, which is needed in the Lyapunov-based stability argument.

Assumption 2.3: The state constraint set $\mathbb{X}$, defined by $\tilde{x} \in \mathbb{X}$, is closed since it is described by linear inequalities. The input constraint set $\mathbb{U}$, defined by $u \in \mathbb{U}$, is bounded by the finite torque limits and is therefore compact. The terminal set $\mathbb{X}_f \subset \mathbb{X}$, taken as $\mathbb{X}_f = \{\tilde{x} \in \mathbb{R}^4 \mid \tilde{x}^\top P \tilde{x} \leq \alpha\}$, is also compact, since it is closed and bounded for $P \succ 0$. Finally, the origin satisfies $0 \in \mathbb{X}$, $0 \in \mathbb{U}$, and $0 \in \mathbb{X}_f$, and because the constraints are symmetric and strictly positive, it lies in the interior of these sets. Therefore, the required properties of Assumption 2.3 are satisfied.

B. Terminal Set Construction

As already stated in Section II, the terminal set is chosen as the ellipsoid $\mathbb{X}_f = \{\tilde{x} \in \mathbb{R}^4 \mid \tilde{x}^\top P \tilde{x} \leq \alpha\}$, where the scalar $\alpha > 0$ determines the size of the set. The matrix P fixes the shape of the ellipsoid through the terminal cost, while α is selected such that the ellipsoid remains admissible with respect to the chosen state and input constraints. In particular, for each constraint of the form $|c^\top \tilde{x}| \leq b$, where $c^\top$ corresponds to the rows of H for state constraints or the feedback gain K for input constraints, the largest admissible value of α is given by

$$\alpha_i = \frac{b^2}{c^\top P^{-1} c}. \tag{15}$$

This expression is applied to the bounds on the first-link angle, the second-link deviation, and the input torque under the local feedback law

$$u = K\tilde{x}, \qquad K = -(R + B_d^\top P B_d)^{-1} B_d^\top P A_d. \tag{16}$$

The final value of α is then chosen as

$$\alpha = \min_i \alpha_i, \tag{17}$$

so that all considered constraints are satisfied simultaneously. For the numerical system considered in this project. Figure 2 shows a two-dimensional slice of the terminal set in the $(\theta_1, \tilde{\theta}_2)$ plane. The grey ellipse represents the corresponding slice of the terminal region, while the dashed lines indicate the state constraints. In addition, the markers show the projected initial state together with several late-time projected states of a closed-loop trajectory. These late-time states lie inside the plotted slice of the terminal region and move toward the origin, illustrating the convergence of the closed-loop response in the neighborhood of the equilibrium.

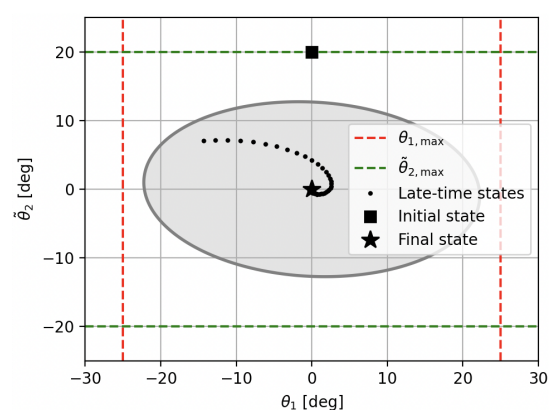

Fig. 2. Two-dimensional slice of the terminal set in the $(\theta_1, \tilde{\theta}_2)$ plane. The grey ellipse shows the terminal region, the dashed lines indicate the imposed state constraints, and the markers show the projected initial state together with several late-time projected states.

C. Terminal Set Invariance and Lyapunov Decrease

Control invariance condition: The terminal set $\mathbb{X}_f$ must be control invariant, meaning that for every state in $\mathbb{X}_f$, there exists an admissible input that keeps the next state inside $\mathbb{X}_f$. This condition is verified using the local terminal controller $u = K\tilde{x}$. Under this feedback law, the closed-loop dynamics are given by

$$\tilde{x}_{k+1} = A_K \tilde{x}_k, \qquad A_K = A_d + B_d K. \tag{18}$$

For the chosen local controller, this means showing that if the state starts in $\mathbb{X}_f$, then the local controller keeps the next state in $\mathbb{X}_f$ as well: $\tilde{x} \in \mathbb{X}_f \Rightarrow A_K \tilde{x} \in \mathbb{X}_f$.

Lyapunov decrease condition: Under the local terminal control, the terminal cost must decrease along the closed-loop dynamics. This is expressed by the condition

$$V_f(A_K \tilde{x}) - V_f(\tilde{x}) \leq -\ell(\tilde{x}, K\tilde{x}), \tag{19}$$

where A_K is defined in Eq. 18. With the terminal cost defined in Eq. 13, substitution gives

$$V_f(A_K \tilde{x}) - V_f(\tilde{x}) = \tilde{x}^\top (A_K^\top P A_K - P)\tilde{x}. \tag{20}$$

Because P is the solution of the discrete-time algebraic Riccati equation associated with the LQR design, it follows that

$$A_K^\top P A_K - P = -(Q + K^\top R K). \tag{21}$$

Substituting Eq 21 into Eq. 20 gives

$$V_f(A_K \tilde{x}) - V_f(\tilde{x}) = -\tilde{x}^\top (Q + K^\top R K)\tilde{x} = -\ell(\tilde{x}, K\tilde{x}). \tag{22}$$

This shows that, under the local terminal controller, the terminal cost decreases at each step. Therefore, $V_f(\tilde{x})$ can be used as a Lyapunov function on the terminal set.

Assumption 2.14: The chosen terminal ingredients satisfy the basic stability assumption. The terminal set $\mathbb{X}_f$ is constructed such that it is admissible with respect to the state and input constraints, and under the local terminal control law $u = K\tilde{x}$ it is positively invariant. Moreover, the terminal cost $V_f(\tilde{x}) = \tilde{x}^\top P \tilde{x}$ satisfies the Lyapunov decrease condition on $\mathbb{X}_f$.

D. Stability for the Nonlinear System

To guarantee asymptotic stability for the true nonlinear system, we verify that the fundamental MPC stability assumptions hold beyond the linearized prediction model. First, the nonlinear dynamics are continuous because the mass matrix $M(q)$ is globally positive definite and therefore everywhere invertible and $f(q, \dot{q}, u)$ is a composition of $\cos(\cdot)$, $\sin(\cdot)$, and polynomial functions, which are all continuous.

Next, we prove the terminal cost acts as a local Lyapunov function. Expressing the nonlinear closed-loop dynamics as a Taylor expansion yields $F(\tilde{x}, K\tilde{x}) = A_K \tilde{x} + e(\tilde{x})$, where the linearization error $e(\tilde{x})$ is bounded by $\mathcal{O}(|\tilde{x}|^2)$. Substituting this into the quadratic terminal cost difference yields:

$$V_f(F(\tilde{x}, K\tilde{x})) - V_f(\tilde{x}) = -l(\tilde{x}, K\tilde{x}) + E(\tilde{x})$$

Crucially, while the nominal descent $-l(\tilde{x}, K\tilde{x})$ is strictly negative and quadratic ($\mathcal{O}(|\tilde{x}|^2)$), the resulting Lyapunov error term $E(\tilde{x})$ is cubic ($\mathcal{O}(|\tilde{x}|^3)$). Because the quadratic descent mathematically dominates the cubic error sufficiently close to the origin, the total energy strictly decreases.

Consequently, there exists an invariant terminal set $\mathbb{X}_{f,nl} = \{x \mid x^T P x \leq \alpha_{nl}\}$ with $\alpha_{nl} \leq \alpha$, where the fractional descent condition $V_f(F(\tilde{x}, K\tilde{x})) - V_f(\tilde{x}) \leq -c \cdot l(\tilde{x}, K\tilde{x})$ holds for $c \in (0, 1)$. By utilizing a bisection algorithm to numerically verify the boundary $x^T P x = \alpha_{nl}$ under the nonlinear dynamics F for a high number of sample points, we "guarantee" that for all states within the restricted nonlinear terminal set $\mathbb{X}_{f,nl} = \{x \mid x^T P x \leq \alpha_{nl}\}$, the terminal cost strictly decreases. This fully satisfies Assumption 2.14 for the true system.

E. Recursive Feasibility and Local Feasibility

Recursive feasibility: Let $\tilde{x}_N^*$ denote the corresponding optimal terminal state. We consider the optimal input sequence at a given time step and shift it one step forward. Suppose that the MPC problem is feasible and that $\tilde{x}_N^* \in \mathbb{X}_f$. Since the terminal set is positively invariant under the local controller $u = K\tilde{x}$, appending $K\tilde{x}_N^*$ to the shifted optimal sequence gives a feasible sequence at the next time step. Thus, once the MPC problem is feasible, it remains feasible along the closed-loop trajectory.

Assumption 2.17: This requires that the terminal set $\mathbb{X}_f$ contains the origin as an interior point. Since $P \succ 0$ and $\alpha > 0$, the chosen terminal set is an ellipsoid with non-empty interior containing the origin. Moreover, because the system is controllable, states sufficiently close to the origin can be steered into the terminal set in a finite number of steps with finite cost. Therefore, the origin lies in the interior of the feasible set $\mathbb{X}_N$, and Assumption 2.17 is satisfied.

F. Stability Conclusion

The previous subsections have verified Assumptions 2.2, 2.3, 2.14, and 2.17 for the considered MPC formulation. Therefore, by Theorem 2.19, the origin of the closed-loop MPC system is asymptotically stable on the feasible set $\mathbb{X}_N$.

In addition, since the terminal cost acts as a Lyapunov function on the terminal set and satisfies the required quadratic bounds, Theorem 2.21 implies exponential stability on the terminal set. Thus, the chosen terminal cost, terminal controller, and terminal set ensure recursive feasibility and closed-loop stability of the MPC controller.

G. Region of Attraction

To obtain a numerical estimate of the region of attraction, the closed-loop MPC controller was simulated over a grid of initial conditions in the $(\theta_1(0), \tilde{\theta}_2(0))$ plane, while the initial velocities were set to zero. An initial condition was classified as belonging to the region of attraction if the MPC problem remained feasible and the state converged to a small neighborhood of the origin. Figure 3 shows the resulting two-dimensional slice of the region of attraction. The controller stabilizes a relatively large set of initial

conditions around the equilibrium, while infeasibility or loss of convergence occurs mainly in the outer corners of the grid, where both initial angular deviations are large. This indicates that the practical region of attraction of the closed-loop MPC controller is significantly larger than the terminal set alone.

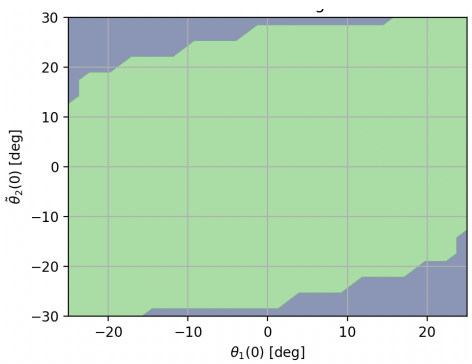

Fig. 3. Numerical estimate of a two-dimensional slice of the region of attraction in the $(\theta_1(0), \tilde{\theta}_2(0))$ plane for zero initial velocities. Green indicates feasible and convergent initial conditions, while blue indicates infeasible or non-convergent cases.

IV. Numerical simulations

In this section, numerical simulations are used to evaluate the performance of the MPC controller for the underactuated double pendulum. The simulations are used to study the influence of the tuning parameters, the prediction horizon, and the imposed constraints, and to compare the final MPC design with a standard LQR controller. To simulate the nonlinear dynamics, the real state update dynamics are updated by solving an initial value problem. The function used is `solve_ivp()` from `scipy.integrate`, using the `RK45` method.

The tuning and horizon studies are first used to determine a suitable set of weighting matrices and a prediction horizon. The resulting controller is then compared with an LQR controller. The full codebase can be found at https://github.com/RaulH03/ModelPredictiveControl.

A. Tuning of Weights (Q and R)

For the tuning study, an initial deviation of $\tilde{\theta}_2(0) = 20°$ is used. This value is large enough to reveal differences between tuning choices, while still representing a local stabilization problem. The controller weights were tuned by varying one parameter at a time while keeping all others fixed at their baseline values. The strongest influence was observed for the weight on the second-link deviation, $q_{\tilde{\theta}_2}$, shown in Figure 4. Increasing $q_{\tilde{\theta}_2}$ reduces the settling time of the passive link, but at the same time increases the overshoot and oscillatory motion of the actuated first link. This behavior is expected, since the passive second link cannot be actuated directly. A larger weight on $\tilde{\theta}_2$ forces the controller to use the actuated first link more aggressively in order to correct the passive-link deviation, which explains the faster recovery of $\tilde{\theta}_2$ and the increased oscillation in θ_1. The value $q_{\tilde{\theta}_2} = 200$ was selected, since it provides the best balance between fast recovery of the passive link and acceptable overshoot in θ_1.

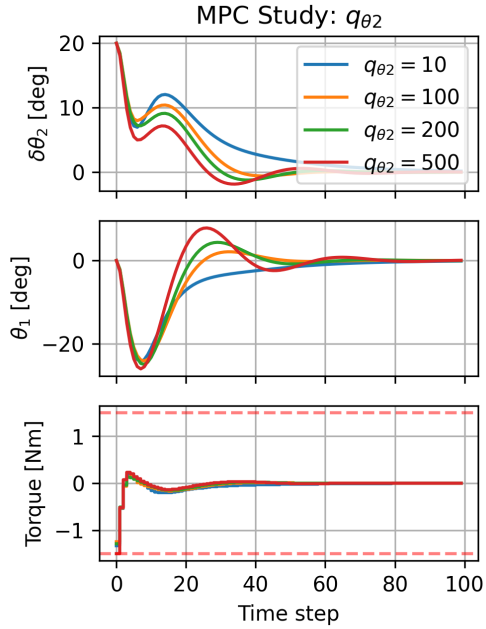

Fig. 4. Effect of varying the weight $q_{\tilde{\theta}_2}$ on the closed-loop response. Larger values improve the regulation of the passive second link, but also lead to a more oscillatory response of the actuated first link.

The velocity weights $q_{\dot{\theta}_2}$ and $q_{\dot{\theta}_1}$ also show a clear trade-off between settling time and damping. As shown in Figure 5, increasing $q_{\dot{\theta}_2}$ leads to a longer settling time of $\tilde{\theta}_2$ and more oscillatory motion in θ_1, so a low value of $q_{\dot{\theta}_2} = 1$ is preferred. Similarly, Figure 6 shows that increasing $q_{\dot{\theta}_1}$ makes the response of both links more sluggish. Although $q_{\dot{\theta}_1} = 0.1$ gives the fastest initial response, it also introduces undesirable oscillations in the first link. Therefore, $q_{\dot{\theta}_1} = 1$ was selected as the best compromise between settling time and well-damped behavior.

The analysis showed that the controller performance is most strongly affected by $q_{\tilde{\theta}_2}$. In contrast, further testing on varying the weight on the first-link angle, q_{θ_1}, and the input weight R produced only minor changes in the closed-loop response. These parameters were therefore kept at their baseline values, namely $q_{\theta_1} = 1$ and $R = 1$.

B. Prediction Horizon Study

Figure 7 compares the closed-loop response for several prediction horizons. The case $N \in \{5, 10\}$ were found to be infeasible and are therefore omitted from the plot, showing that such a short horizon does not provide enough look-ahead for the controller to satisfy the constraints and steer the predicted trajectory toward the terminal region. The horizon $N = 15$ gives a feasible response, but is squished compared to the longer horizons. This is caused by the terminal constraint being forced earlier, demanding more aggressive control. By contrast, the horizons from $N = 20$ onward produce nearly identical trajectories and control inputs, since the controller can better anticipate the coupled motion of both links over a longer time window. Since $N = 20$ gives similar performance as $N = 25$ with a smaller optimization problem, it is selected as the final horizon.

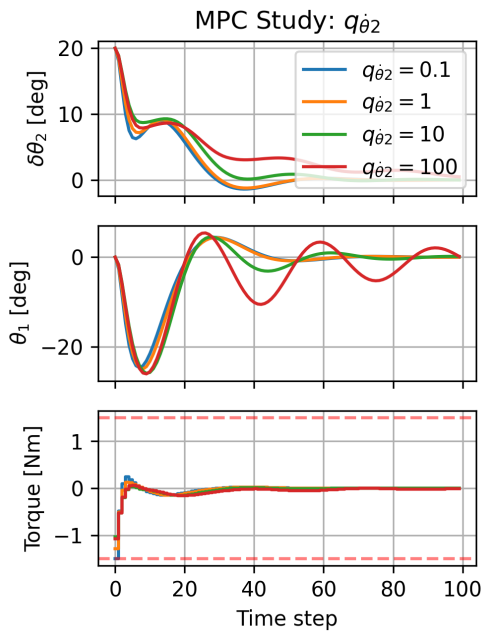

Fig. 5. Effect of varying the weight $q_{\dot\theta_2}$ on the closed-loop response. Increasing $q_{\dot\theta_2}$ leads to slower settling of the passive second-link deviation $\tilde\theta_2$ and larger overshoot in the actuated first-link angle θ_1.

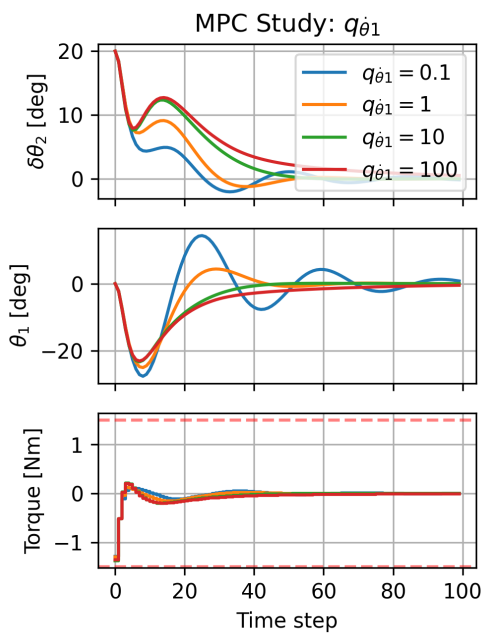

Fig. 6. Effect of varying the weight $q_{\dot\theta_1}$ on the closed-loop response. Increasing $q_{\dot\theta_1}$ slows down the regulation of the passive second-link deviation $\tilde\theta_2$ and makes the response of the actuated first link θ_1 more sluggish.

C. Constraint Analysis

To study the effect of the constraints more directly, the tuned MPC controller was simulated for increasing initial deviations of the passive link, $\tilde\theta_{2,0} \in \{20°, 22°, 24°, 26°, 28°, 30°\}$. Over this range, the dominant active constraint is the input bound, while the state constraints remain inactive. Figure 8 shows the corresponding control input for representative initial

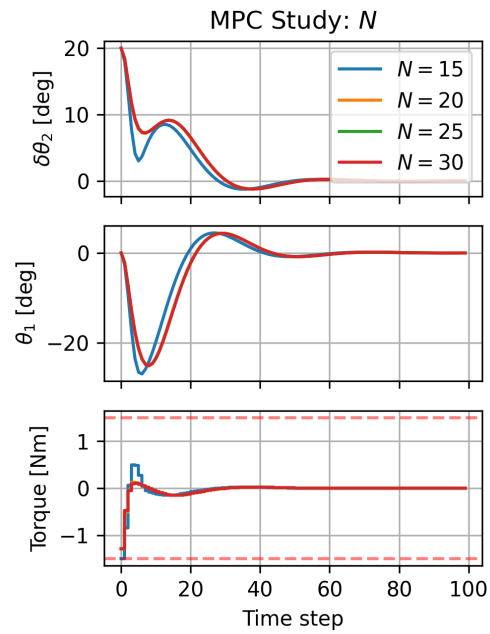

Fig. 7. Closed-loop response for different prediction horizons. The top and middle plots show the state trajectories $\tilde\theta_2$ and θ_1, while the bottom plot shows the corresponding control input u.

conditions. For $\tilde\theta_{2,0} \in \{20°, 22°\}$ the input remains within the admissible range and the response is smooth. At $\tilde\theta_{2,0} = 24°$, the input constraint becomes active for the first time, marking the onset of constraint-limited operation. For larger initial deviations, up to $\tilde\theta_{2,0} = 30°$, the controller keeps the problem feasible, but the control input exhibits a clearly stronger initial peak and a less smooth response due to the active constraint. Increasing the initial deviation further to $\tilde\theta_2(0) = 32°$ makes the MPC problem infeasible. This indicates that, for the selected tuning and horizon, the input constraint starts to affect the response at around $24°$, while feasibility is retained up to approximately $32°$ passive-link deviation.

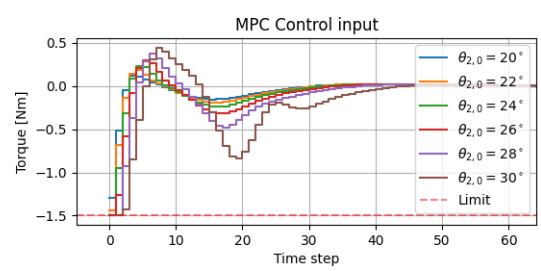

Fig. 8. Zoomed in plot of control input u for different initial deviations of the passive-link state. The dashed red lines indicate the actuator limits $\pm u_{\max}$. For $\tilde\theta_{2,0} \in \{20°, 22°\}$, the input remains within the admissible range, while at $\tilde\theta_2(0) = 24°$ the controller first reaches the input constraint.

D. Baseline Comparison (MPC vs LQR)

For an initial deviation of $\tilde\theta_{2,0} = 20°$, the MPC and LQR controllers produce very similar responses, since the imposed constraints remain inactive in this case. To better highlight

the effect of explicit constraint handling, the comparison is therefore carried out for a larger initial deviation of $\tilde{\theta}_{2,0} = 30°$, where the input constraint are active. Figure 9 shows the corresponding control input for both controllers.

While the MPC input remains within the admissible range, the unconstrained LQR controller immediately asks for a torque that exceeds the actuator limit. In a real system, the motor would simply not be able to produce this input. This shows the main advantage of MPC in the present setting: the available motor torque is included directly in the optimization problem, so the computed control action remains physically feasible even for larger initial deviations.

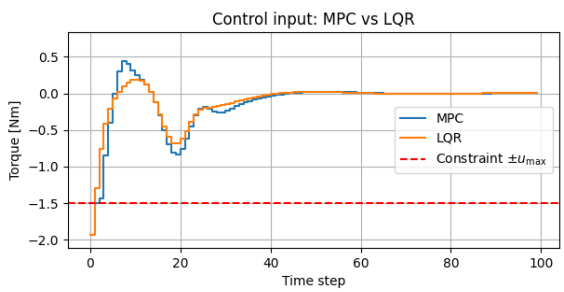

Fig. 9. Control input for MPC and LQR with an initial deviation of $\tilde{\theta}_{2,0} = 30°$. The dashed red lines indicate the actuator limits $\pm u_{\max}$. While the MPC controller respects the input bound, the unconstrained LQR controller demands a torque beyond the admissible range.

E. Measurement Noise

In practical applications, sensors are rarely perfect and often introduce small errors into the measured state. To evaluate the robustness of our MPC controller, we simulate these inaccuracies by adding random Gaussian noise to the state measurements before they are passed to the optimizer.

Instead of the controller receiving the true state x_k, it receives a noisy version $\hat{x}_k$, defined by

$$\hat{x}_k = x_k + v_k, \tag{23}$$

where x_k is the true state of the system at time step k, and v_k is the measurement noise, modeled as a random variable following a normal distribution $v_k \sim \mathcal{N}(0, \Sigma)$. The noisy state $\hat{x}_k$ is then used by the MPC controller to compute the next control input.

To show the effect of measurement noise, the controller is tested for a relatively high noise level with standard deviations of $0.5°$ in θ_1, $1.0°$ in $\tilde{\theta}_2$, and $2.0°$ in both velocity states. Figure 10 compares the noise-free and noisy responses. The controller still stabilizes the initial offset and keeps the pendulum close to the equilibrium. The main visible effect of the noise is a small fluctuation of the state trajectories around zero and a noisier control input near the equilibrium. Overall, the results indicate that the proposed MPC controller remains robust for this measurement noise level.

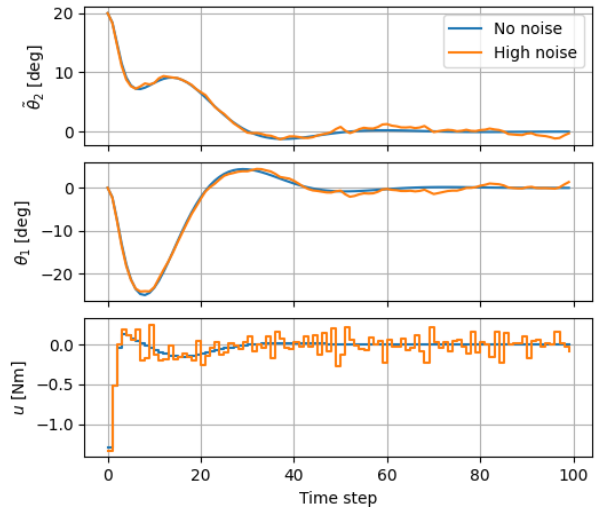

Fig. 10. Closed-loop response of the MPC controller with and without measurement noise. The top and middle plots show the trajectories of $\tilde{\theta}_2$ and θ_1, while the bottom plot shows the corresponding control input.

V. Conclusions

In this project an MPC controller was designed that successfully stabilized an underactuated two-link pendulum while staying under a 1.5 Nm torque limit on the actuator. Comparing the controller to a conventional LQR controller, the designed controller can stabilize the pendulum with higher offsets from the equilibrium while staying under this torque limit. A proof is included for the asymptotic stability of the controller using a DARE based terminal cost and invariant terminal set. The final controller also demonstrated robustness to measurement noise.

During tuning a trade off had to be made between the fast recovery of the second link and the overshoot of the first link. A faster recovery would use a higher input and had a higher amplitude on the first link. A balance was struck between recovery speed and actuation of the system. During a study on the prediction horizon a horizon on $N20$ was found to be sufficient for this problem, with a longer horizon not providing any benefit.

The primary limitation of the current controller is the reliance on the local linear prediction model. Deviations far from the equilibrium cause the non linear dynamics to diverge from the linear model. This divergence, combined with the torque limit imposed, restrict the feasible starting condition to a initial deviation of $30°$ of the passive link.

A next logical step would be implementing a nonlinear MPC strategy. Using the full nonlinear system would eliminate the limitations of the locally linearized system and likely enlarge the region of attraction. Another extension that could be made is adding an observer, alongside an explicit disturbance rejection mechanism into the system. This could make the system more robust to noise.

References

[1] J. Rawlings and D. Mayne, *Model Predictive Control: Theory and Design*. Nob Hill Publishing, 2008.